FLIGHT DELAY PREDICTION USING MACHINE LEARNING

Bachelors of Technology in Computer Science Engineering

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ABSTRACT

This project builds an end-to-end machine learning system that predicts whether a U.S. domestic flight will arrive 15 or more minutes late, using only information available 24 hours before departure. The system employs a LightGBM gradient-boosted decision tree trained on the Bureau of Transportation Statistics (BTS) On-Time Performance dataset (~7 million flights). A 17-feature engineering pipeline transforms raw records through cyclic temporal encoding, Haversine great-circle distance, K-Fold smoothed target encoding (λ=20), historical delay aggregates, and holiday proximity detection.

A strict temporal split (Jan–Aug training, Sep–Oct validation, Nov–Dec test) simulates real deployment. Evaluation uses PR-AUC as the primary metric given the ~20% imbalanced positive class. A three-stage leakage audit (blacklist enforcement, high-correlation scan, SHAP importance check) ensures no post-departure information contaminates predictions. SHAP TreeExplainer provides per-prediction feature attributions.

The model is deployed as a Streamlit web application on Streamlit Community Cloud, featuring an interactive 3D globe, airline-branded boarding pass cards, a SHAP waterfall chart, and a modular component architecture. A FastAPI + React alternative is also containerised via Docker Compose.

1. Introduction

Flight delays cost the U.S. economy over $30 billion annually. In 2023, approximately 21% of domestic flights arrived more than 15 minutes late. While airlines monitor delays internally, passengers and operational planners lack accessible pre-departure predictive tools—existing consumer tools primarily report real-time status rather than forecasts.

From a machine learning perspective, flight delay prediction presents a moderately imbalanced binary target (~20% delayed), rich categorical features (airline, airport, aircraft), temporal patterns (hour, day, season), and geospatial structure (route distance). Critically, it also introduces **temporal information leakage**: many BTS columns (taxi times, actual departure/arrival, delay breakdowns) are only available post-departure. Using these features produces artificially inflated accuracy that is useless in practice.

This project addresses this by enforcing a strict T-24h information boundary. Every feature—from cyclic departure-time encoding to smoothed historical delay rates—is knowable 24 hours before departure. The system spans the full ML lifecycle: data ingestion, temporal splitting, feature engineering, Bayesian hyperparameter optimisation, SHAP interpretability, leakage auditing, and deployment as a full-stack web application.

2. Problem Statement

> ***Given only information available 24 hours before a scheduled departure—airline, origin airport, destination airport, scheduled departure time, and aircraft tail number—can a machine learning model accurately predict whether the flight will arrive 15 or more minutes late?***

The task is a supervised binary classification problem:

- **Label:** `y = 1` if arrival delay ≥ 15 minutes; `y = 0` otherwise
- **Positive class (~20%):** Delayed | **Negative class (~80%):** On-time

The model must satisfy four domain-specific requirements:

1. **Strict temporal leakage prevention:** A blacklist of 14 post-departure columns enforced at load time, plus a SHAP-based sanity check (no single feature >40% total importance).
2. **Appropriate evaluation for class imbalance:** PR-AUC as primary metric (a naïve "always on-time" classifier achieves ~80% accuracy but is useless).
3. **Interpretable predictions:** Per-prediction top contributing factors with signed SHAP values.
4. **Temporal evaluation integrity:** Chronological train/val/test split—no random shuffling.

3. Literature Review

Rebollo & Balakrishnan (2014) applied Random Forests to U.S. airport departure delays, demonstrating that tree-based ensembles outperform linear models by capturing non-linear interactions between weather, traffic, and temporal features.

Kim et al. (2016) compared Logistic Regression, SVM, Random Forest, and Gradient Boosting on BTS data, finding Gradient Boosting consistently achieved the highest AUC. They emphasised the critical importance of removing post-departure features to avoid information leakage.

Ke et al. (2017) introduced LightGBM, a gradient-boosting framework using histogram-based splitting and leaf-wise growth, achieving state-of-the-art tabular performance while training 10–20× faster than traditional GBDT implementations.

Akiba et al. (2019) introduced Optuna, a hyperparameter optimisation framework using Tree-structured Parzen Estimators (TPE) that converges to near-optimal configurations in 30–50 trials on typical search spaces.

Lundberg & Lee (2017) introduced SHAP, providing a theoretically grounded framework for feature attribution. TreeSHAP computes exact Shapley values in polynomial time for tree models, enabling per-prediction explanations at production speed.

4. Dataset Description

4.1 Source & Scale

The **BTS On-Time Performance** dataset from the U.S. Bureau of Transportation Statistics covers all U.S. carriers with ≥0.5% of domestic passenger revenue. A typical full-year dataset contains ~6–7 million flight records across 15+ carriers and 300+ airports. Available at `https://www.transtats.bts.gov/`.

4.2 Required Fields

Column	Type	Description
`FL_DATE`	date	Flight date (YYYY-MM-DD)
`OP_CARRIER`	string	IATA airline code (e.g., "AA", "DL")
`ORIGIN`, `DEST`	string	Airport IATA codes
`CRS_DEP_TIME`	integer	Scheduled departure (HHMM format)
`CRS_ELAPSED_TIME`	float	Scheduled duration (minutes)
`DISTANCE`	float	Route distance (statute miles)
`ArrDelay`	float	Arrival delay in minutes
`TAIL_NUM`	string	Aircraft tail number
`Month`, `DayOfWeek`	integer	Calendar month / ISO day of week

4.3 Target Variable

$$\mathbf{y} = 1 \text{ if ArrDelay} \geq 15 \text{ minutes (Delayed)} \quad | \quad \mathbf{y} = 0 \text{ otherwise (On-time)}$$

4.4 Blacklisted Columns (14 Unique, 26 Including Case Variants)

Blacklisted Column	Reason
`TaxiOut`, `TaxiIn`	Only known after pushback / landing

`WheelsOff`, `WheelsOn`	Actual takeoff / landing times
`ActualElapsedTime`, `AirTime`	Only computable post-flight
`CarrierDelay`, `WeatherDelay`, `NASDelay`, `SecurityDelay`, `LateAircraftDelay`	Post-flight delay attribution
`DiversionDistance`	Only applies to diverted flights
`ActualDeparture`, `ActualArrival`	Post-departure times

5. Data Preprocessing & Temporal Splitting

5.1 Loading Pipeline

The `load_raw()` function accepts CSV/Parquet, standardises BTS column name variants, enforces the 26-string blacklist (14 columns × 2 case variants), creates the binary label, drops cancelled/diverted rows (null `ArrDelay`), and sorts chronologically by `FL_DATE`.

5.2 Temporal Split Strategy

Split	Months	Approx. Size	Purpose
Train	January–August (1–8)	~70%	Model learning
Validation	September–October (9–10)	~15%	Hyperparameter tuning, threshold selection
Test	November–December (11–12)	~15%	Final held-out evaluation

Unlike random splits, temporal splitting simulates real deployment where the model only predicts future flights from past data. Runtime assertions enforce strict chronological ordering:

```
assert max(train_dates) < min(val_dates),   "Train/val date overlap detected"
assert max(val_dates)   < min(test_dates),  "Val/test date overlap detected"
```

6. Feature Engineering

The `FeaturePipeline` class in `src/features/pipeline.py` transforms raw flight records into 17 predictive features through five sequential stages.

6.1 Cyclic Temporal Encoding

Raw time variables are inherently circular (23:59 is adjacent to 00:01). Sin/cos encoding maps each value to the unit circle:

$$x_{sin} = \sin(2\pi \cdot t / T) \qquad x_{cos} = \cos(2\pi \cdot t / T)$$

Output Features	Input	Period (T)
`CRS_DEP_TIME_sin/cos`	Minutes since midnight	1440
`DayOfWeek_sin/cos`	Day of week (1–7)	7
`Month_sin/cos`	Month (1–12)	12
`departure_hour`	Hour (0–23)	— (ordinal)

6.2 Haversine Great-Circle Route Distance

$$a = \sin^2(\Delta lat / 2) + \cos(lat_1) \cdot \cos(lat_2) \cdot \sin^2(\Delta lon / 2)$$

$$distance = 2 \times R \times \arcsin(\sqrt{a}) \quad \text{where } R = 6{,}371.0088 \text{ km}$$

Vectorised NumPy implementation using coordinates for **139 major U.S. airports**. Fallback: BTS `DISTANCE` × 1.60934 km if coordinates are missing; `NaN` (handled natively by LightGBM) if neither is available.

6.3 Historical Delay Rate Aggregates

Route Delay Rate (min. 30 flights)

$$w = \min(n, 30) / 30 \qquad route_delay_rate = w \cdot (n_{delayed} + 1) / (n_{total} + 2) + (1 - w) \cdot p_{global}$$

Airline Delay Rate (min. 100 flights)

$$w = \min(n, 100) / 100 \qquad \text{airline_delay_rate} = w \cdot (n_{delayed} + 1) / (n_{total} + 2) + (1 - w) \cdot p_{global}$$

Laplace smoothing (+1 / +2) prevents extreme rates. Both computed on training data only; applied to val/test via lookup.

6.4 K-Fold Smoothed Target Encoding

High-cardinality categoricals (ORIGIN: 300+ airports, TAIL_NUM: 5,000+ aircraft) are target-encoded with K=5 out-of-fold encoding to prevent target leakage:

$$TE(\text{category}) = (n \cdot p_{category} + \lambda \cdot p_{global}) / (n + \lambda) \quad \text{where } \lambda = 20.0$$

Columns encoded: `OP_CARRIER` → `OP_CARRIER_te`, `ORIGIN` → `ORIGIN_te`, `DEST` → `DEST_te`, `TAIL_NUM` → `TAIL_NUM_te`. Unseen categories at inference receive the global mean.

6.5 Holiday Proximity

$$\texttt{is_near_holiday} = 1 \text{ if } \min(|\text{flight_date} - \text{holiday_date}|) \leq 2 \text{ days, else } 0$$

Ten U.S. federal holidays detected (4 fixed-date + 6 variable-date). Vectorised with chunked broadcasting (50k rows/chunk).

6.6 Complete Feature Set (17 Features)

#	Feature	Stage
1–2	`CRS_DEP_TIME_sin/cos`	Cyclic encoding
3–4	`DayOfWeek_sin/cos`	Cyclic encoding
5–6	`Month_sin/cos`	Cyclic encoding
7	`departure_hour`	Extracted hour
8	`route_distance_km`	Haversine / fallback
9	`DISTANCE`	Raw BTS (statute miles)
10	`CRS_ELAPSED_TIME`	Raw BTS (minutes)
11–14	`OP_CARRIER_te`, `ORIGIN_te`, `DEST_te`, `TAIL_NUM_te`	Target encoding
15–16	`route_delay_rate`, `airline_delay_rate`	Historical aggregates
17	`is_near_holiday`	Holiday flag

7. LightGBM & Hyperparameter Optimisation

7.1 Why LightGBM?

Criterion	Advantage
Mixed features	Handles numeric + encoded categoricals natively
Missing values	Built-in NaN handling (assigns to best split direction)
Speed	Histogram-based splitting: 10–20× faster than traditional GBDT
Imbalance	Native `scale_pos_weight`
Interpretability	Compatible with SHAP TreeExplainer

Key innovations: histogram-based splitting (255 bins), leaf-wise (best-first) growth, Gradient-based One-Side Sampling (GOSS), and Exclusive Feature Bundling (EFB).

7.2 Default Parameters

```
objective: binary    learning_rate: 0.03    num_leaves: 127
max_depth: -1        min_child_samples: 50  feature_fraction: 0.8
bagging_fraction: 0.8    bagging_freq: 5    lambda_l1: 0.1    lambda_l2: 1.0
max_rounds: 3,000    early_stopping: 100 rounds
```

7.3 Optuna Bayesian Optimisation

Hyperparameter	Range	Scale
`num_leaves`	20–150	Integer
`max_depth`	4–12	Integer
`min_child_samples`	10–100	Integer
`feature_fraction`	0.5–0.9	Continuous
`bagging_fraction`	0.6–0.9	Continuous
`learning_rate`	0.01–0.1	**Log-uniform**
`lambda_l1`, `lambda_l2`	0.0–5.0	Continuous

Protocol: TPESampler (seed=42), 50 trials, maximising validation PR-AUC. Per trial: up to 2,000 rounds with early stopping (patience=50).

8. Model Training Process

8.1 End-to-End Data Flow

```
Raw BTS CSV
  |-- load_raw()          --> cleaned DataFrame (blacklist enforced, y created)
  |-- temporal_split()    --> train (Jan-Aug), val (Sep-Oct), test (Nov-Dec)
  |
  |-- FeaturePipeline.fit_transform(train)   --> X_train (17 features)
  |-- FeaturePipeline.transform(val/test)    --> X_val, X_test
  |
  |-- train_lgbm(X_train, y_train, X_val, y_val)  --> LightGBM Booster
  |-- save_model()  --> model_YYYYMMDD_HHMMSS.txt + params.json
```

8.2 Threshold Selection

The optimal decision threshold maximises F1-score on the validation set rather than using the default 0.5:

$$F1(t) = 2 \cdot \text{precision}(t) \cdot \text{recall}(t) \,/\, (\text{precision}(t) + \text{recall}(t))$$

$$\text{optimal_threshold} = \text{argmax}_t \, F1(t)$$

8.3 Artifacts Saved

File	Contents
`model_YYYYMMDD_HHMMSS.txt`	Trained model (tree structure + weights)
`params.json`	Threshold, validation metrics, hyperparameters
`pipeline.pkl`	Fitted FeaturePipeline (all encoders + aggregates)
`target_encoder.pkl`	Fitted KFoldTargetEncoder
`best_params.json`	Optuna best hyperparameters

9. Evaluation & Results

9.1 Why Not Just Accuracy?

With ~20% positive rate, a trivial “always on-time” classifier gets ~80% accuracy. PR-AUC focuses on the model’s ability to correctly identify the minority delayed class, making it far more informative.

9.2 Metric Suite

Metric	Description
PR-AUC	Primary: area under precision–recall curve
ROC-AUC	Receiver operating characteristic AUC
F1-Score	Harmonic mean of precision & recall at chosen threshold
Precision @ 50% Recall	Precision when catching half of all delays
Recall @ 80% Precision	Delays caught when 80% of predictions are correct
Brier Score	Calibration error (lower is better)

9.3 Test-Set Results

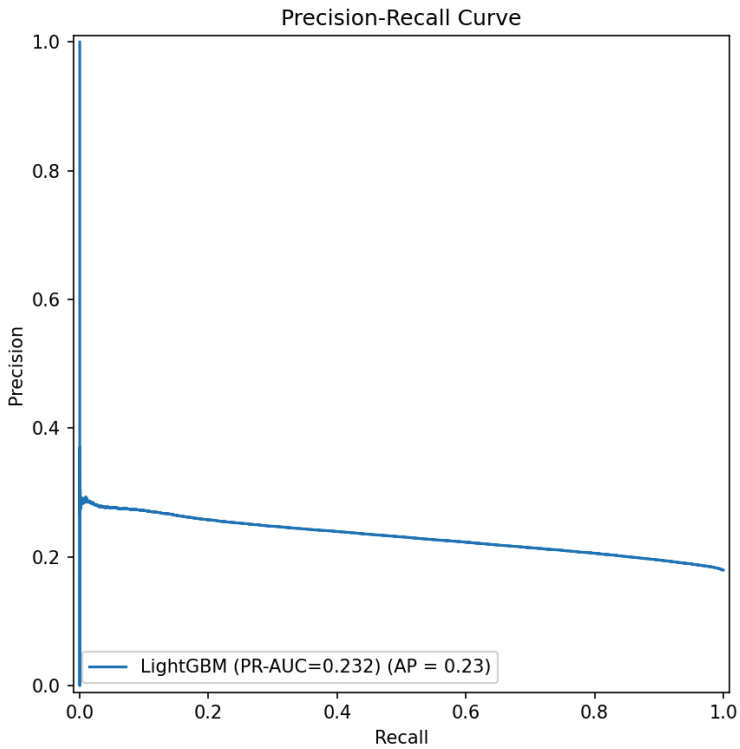


Figure 1: *Precision–Recall curve on held-out Nov–Dec test set*

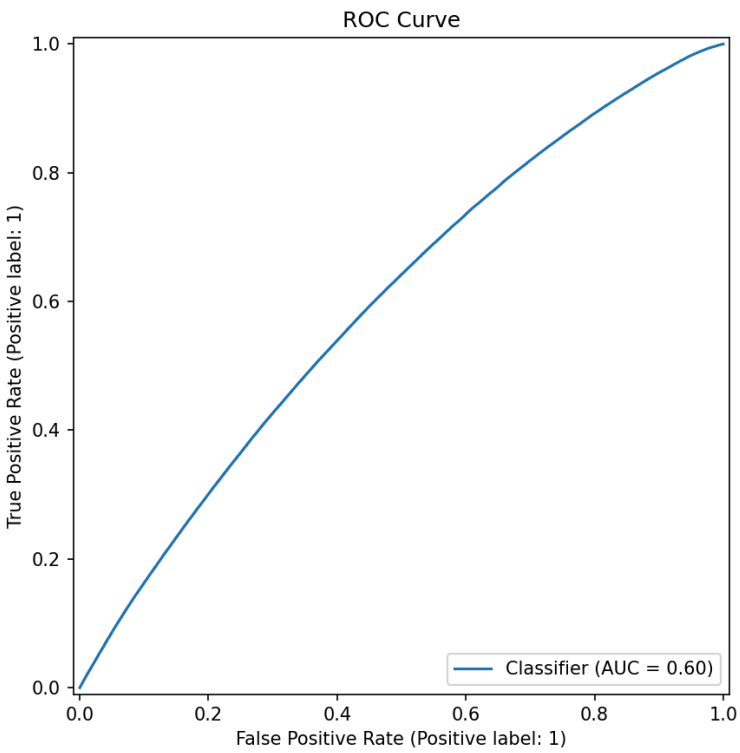


Figure 2: *ROC curve on held-out Nov–Dec test set*

9.4 Confusion Matrix

False Negatives (delayed flights predicted on-time) carry higher cost than **False Positives** (on-time flights predicted delayed). The `scale_pos_weight` parameter and

threshold tuning both shift the model toward higher recall when desired.

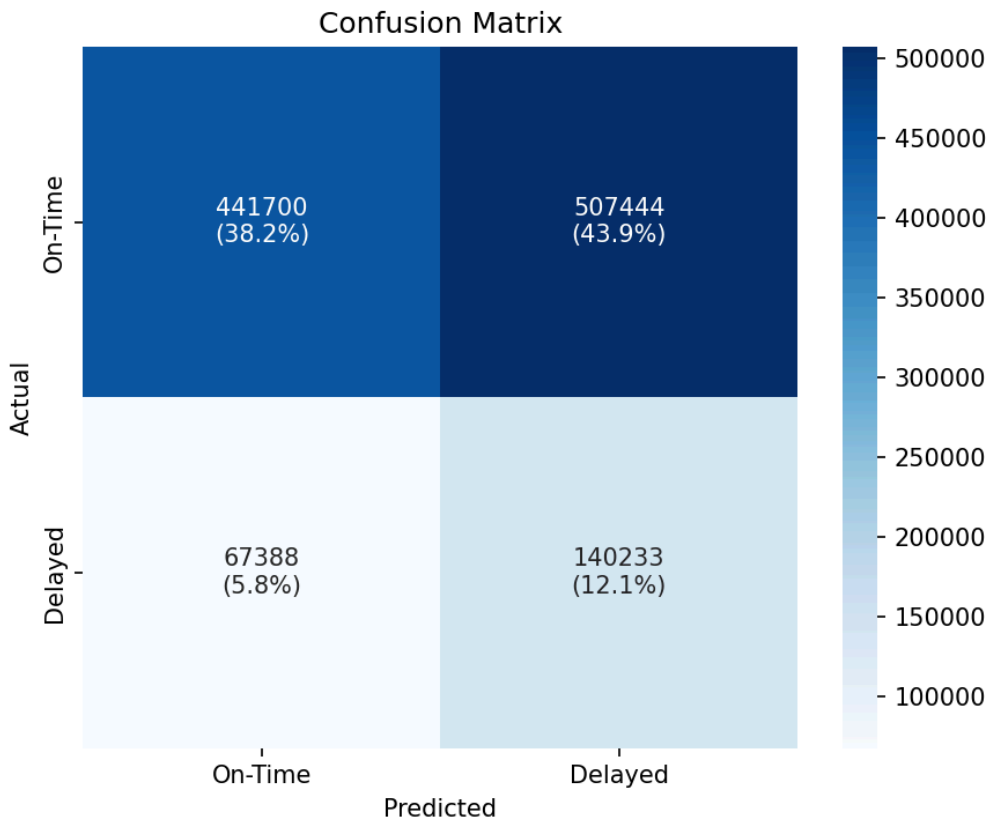


Figure 3: *Confusion matrix on the held-out test set*

9.5 Calibration

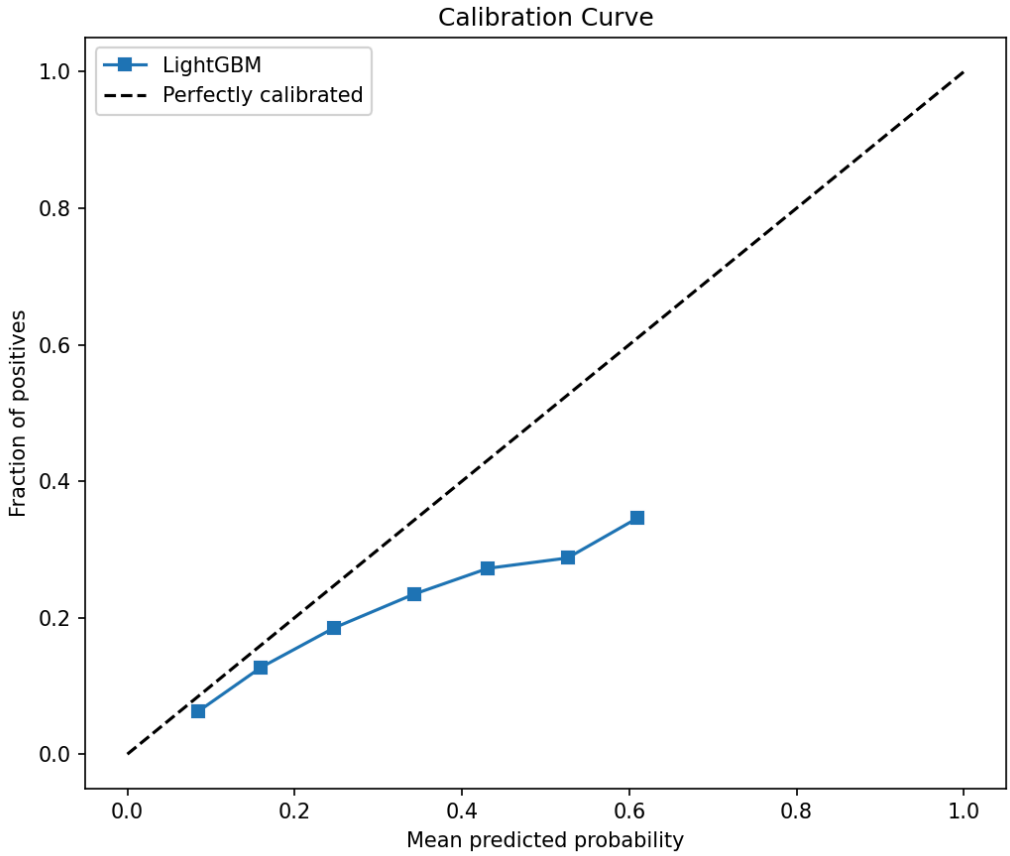


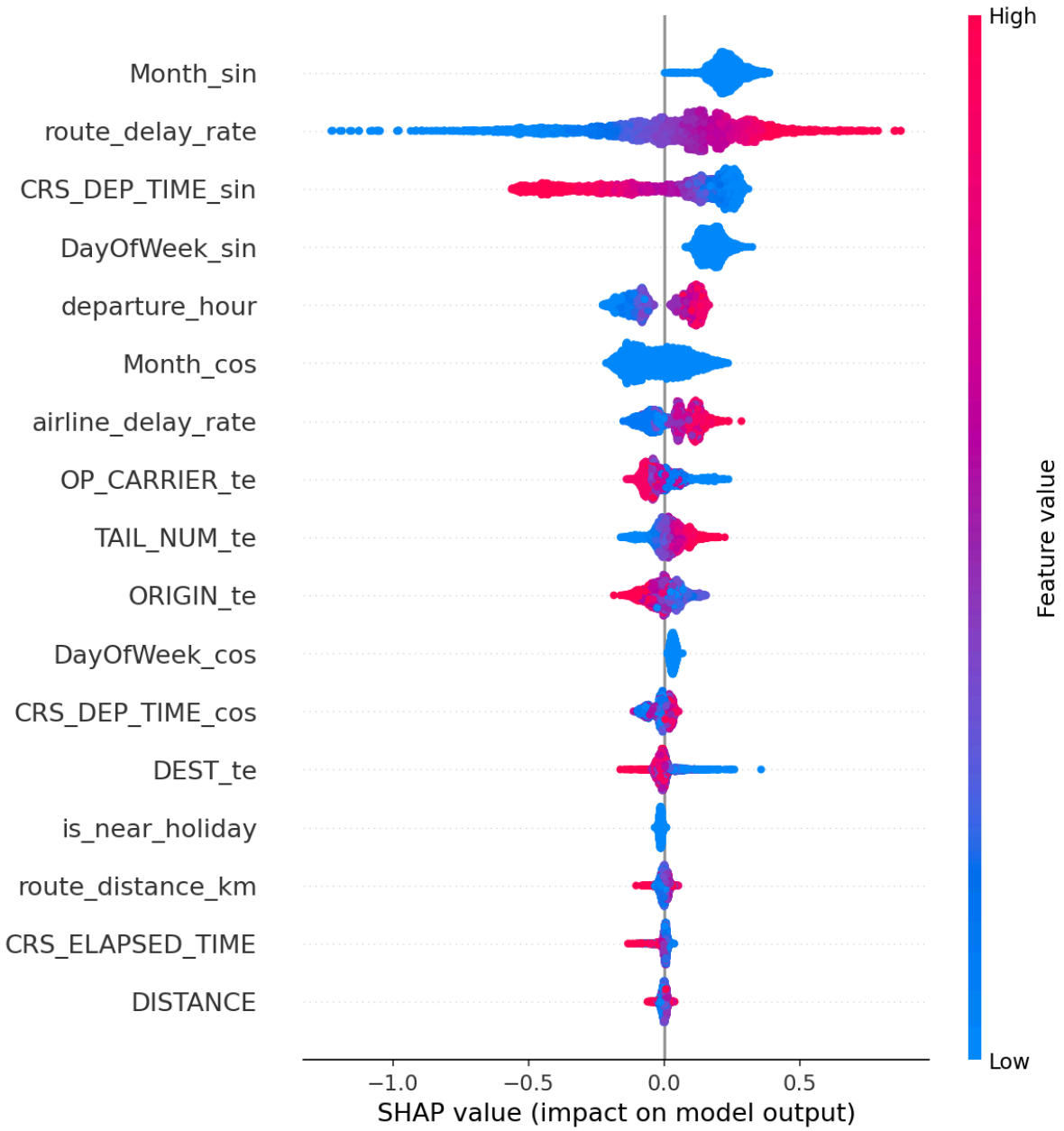
Figure 4: *Calibration curve: predicted probability vs. observed frequency*

10. SHAP Interpretability & Leakage Audit

10.1 Global Feature Importance

SHAP TreeExplainer applied to 2,000 random test-set rows produces two complementary views:

1. **Beeswarm:** Each point = one flight. Horizontal position = SHAP value. Colour = feature value (red=high, blue=low). Shows both importance and direction of effect.
2. **Bar:** Mean |SHAP| per feature—a simple importance ranking.



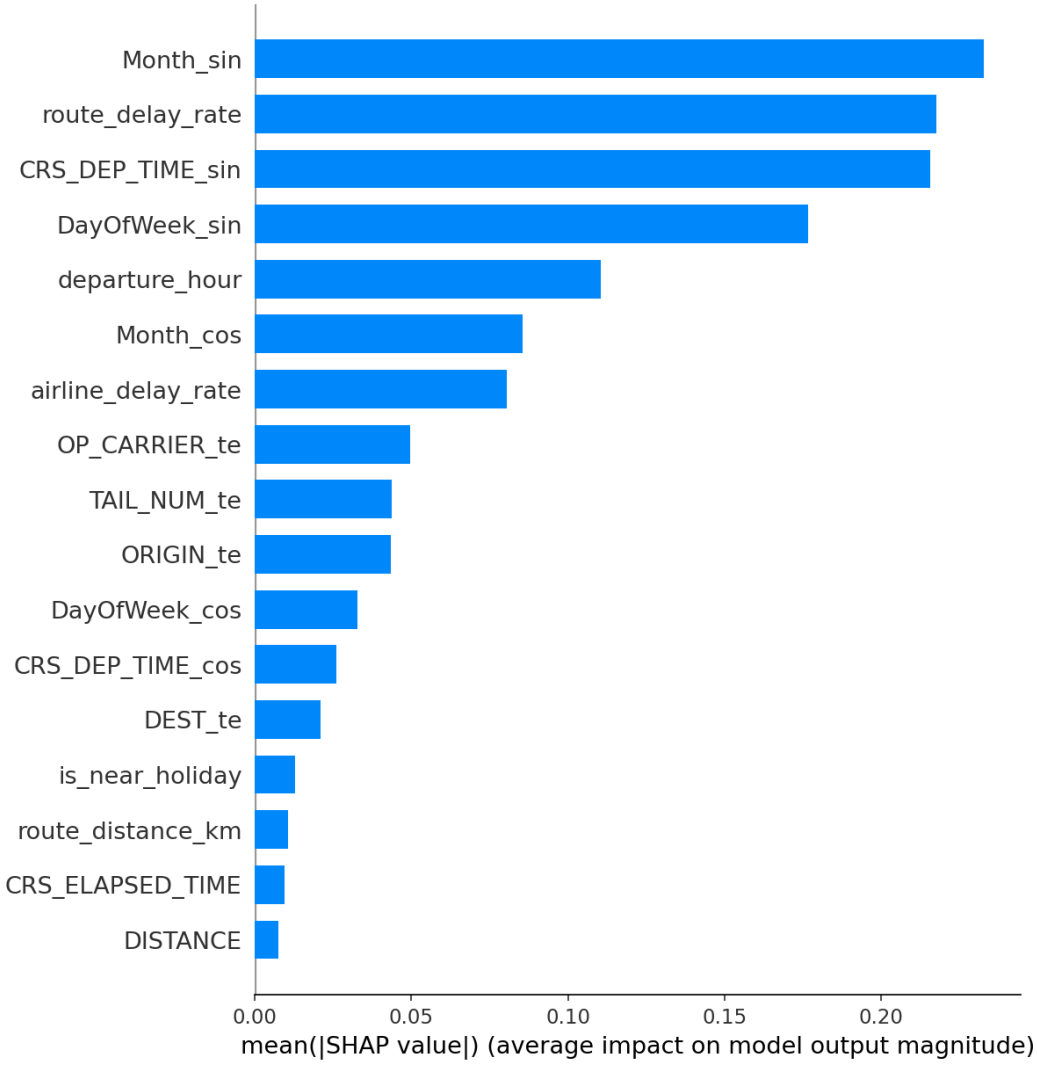


Figure 5: *SHAP global feature importance: beeswarm and mean |SHAP| bar chart*

10.2 Per-Prediction Explanations

Each prediction returns the top 5 SHAP factors with signed impacts:

```
"top_factors": [
    {"feature": "airline_delay_rate", "impact": +0.1523},
    {"feature": "departure_hour",     "impact": +0.1198},
    {"feature": "route_distance_km",  "impact": -0.0812},
    {"feature": "ORIGIN_te",          "impact": +0.0534},
    {"feature": "Month_sin",          "impact": -0.0291}
]
```

10.3 Three-Stage Leakage Audit

Stage	Check	Threshold
1	No blacklisted column present in data	Any match → `ValueError`

2	Pearson correlation with target	|r| > 0.95 → flagged
3	SHAP importance concentration	Any feature > 40% → warning

```
============================================================
  LEAKAGE AUDIT REPORT
============================================================
[PASS] Blacklist check: no leakage columns found.
[PASS] Correlation check: no feature exceeds |r| = 0.95.
------------------------------------------------------------
  Overall: ALL CHECKS PASSED
============================================================
```

11. Web Application & Deployment

11.1 Architecture

```
+-----------------+   POST /predict   +------------------+
|   React UI      | ----------------> |  FastAPI         |
|   (Port 3000)   | <---------------- |  (Port 8000)     |
+-----------------+   JSON response   +--------+---------+
                                              |
                                     +--------v---------+
                                     |  LightGBM Model  |
                                     |  + SHAP Values   |
                                     +------------------+
```

11.2 FastAPI Backend

Method	Path	Description
`GET`	`/health`	Model status, load state, version
`POST`	`/predict`	Flight details → prediction + SHAP factors

Pydantic v2 request validation, CORS middleware, request logging. The `FlightDelayModel` singleton auto-discovers the latest model and pipeline from the artifacts directory and initialises SHAP TreeExplainer during startup.

11.3 Streamlit Cloud Deployment

The primary production deployment is a **Streamlit** application deployed on Streamlit Community Cloud. It reuses the same LightGBM model and feature pipeline in-process, with a modular component architecture (`config.py`, `core.py`, `components/` package). Features include:

- Interactive 3D globe (globe.gl + Three.js) with animated route arcs
- Airline-branded boarding pass cards with logo and accent colours
- SHAP waterfall chart in log-odds space with sigmoid probability labels
- Semi-circle probability gauge with colour-coded risk assessment

- Step-by-step `st.status()` loading feedback

Live: `https://flight-delay-predictor-y7veehaanvd2umrlc4z4zh.streamlit.app/`

11.4 Docker Compose

Backend (Python + Uvicorn, port 8000) and frontend (React → Nginx, port 3000) containerised with health checks, read-only artifact volumes, and a shared bridge network. Single-command deployment: `docker-compose up --build`.

12. Project Structure

```
flight-delay-predictor/
├── notebooks/
│   ├── 01_eda.ipynb                  # Exploratory data analysis
│   ├── 02_feature_engineering.ipynb  # Feature pipeline visualisation
│   ├── 03_training.ipynb             # Model training + threshold selection
│   └── 04_evaluation.ipynb           # Evaluation, SHAP, leakage audit
├── src/
│   ├── data/
│   │   ├── loader.py                 # BTS loading + blacklist enforcement
│   │   └── splitter.py               # Temporal train/val/test split
│   ├── features/
│   │   ├── cyclic.py                 # Sin/cos temporal encoding
│   │   ├── geospatial.py            # Haversine distance (139 airports)
│   │   ├── target_encoding.py       # K-Fold smoothed target encoding
│   │   └── pipeline.py              # FeaturePipeline orchestrator
│   ├── model/
│   │   ├── train.py                  # LightGBM training + save
│   │   ├── evaluate.py              # Metrics, SHAP, calibration plots
│   │   └── optimize.py              # Optuna hyperparameter tuning
│   └── utils/
│       └── leakage_check.py         # Three-stage leakage audit
├── backend/                          # FastAPI inference server
├── frontend/                         # React + Vite + Tailwind UI
├── streamlit_app.py                  # Streamlit Cloud deployment
├── config.py                         # Constants, airline/airport data
├── core.py                           # Model loading, inference, utilities
├── components/                       # Modular Streamlit UI components
├── docker-compose.yml
└── requirements.txt
```

13. Design Decisions

Decision	Rationale
LightGBM over deep learning	Handles mixed features natively, SHAP-compatible, fast training. Deep learning offers less interpretability for tabular tasks.
PR-AUC as primary metric	More informative than ROC-AUC on ~20% imbalanced positive class.
Temporal split (no shuffle)	Simulates real deployment. Random splits produce optimistic estimates.
K-Fold target encoding (λ=20)	Prevents target leakage inherent in naïve mean encoding. Smoothing regularises rare categories.
14-column blacklist at load time	Defence-in-depth: leakage blocked at earliest possible stage.
SHAP per prediction	Users need to understand *why*, not just the probability. Builds trust.
Threshold tuning on validation	Enables precision–recall trade-off: stakeholders choose their operating point.
Docker Compose	Single-command deployment with health checks and read-only volumes.

14. Limitations & Future Scope

14.1 Current Limitations

- **No real-time weather:** Weather causes ~30% of delay minutes but airport-specific 24h forecasts lack sufficient accuracy. The model captures weather only indirectly via seasonal patterns and historical rates.
- **U.S. domestic only:** Trained exclusively on BTS data; does not generalise to international routes or non-U.S. carriers.
- **No operational state:** Cannot account for active ground stops, runway closures, or cascading delays—only available minutes before departure.
- **Binary simplification:** Predicts delayed/on-time only, not delay magnitude. Regression could address this.
- **Cold start:** New airlines/airports/aircraft receive the global mean delay rate until sufficient data accumulates.

14.2 Future Scope

- **Weather integration:** NOAA/OpenWeatherMap API forecasts at origin and destination airports.
- **Delay propagation:** Real-time aircraft delay state and airport congestion for a T-2h prediction window.
- **Multi-year training:** 3–5 years of BTS data for robustness to seasonal variation.
- **Quantile regression:** Predict delay magnitude with uncertainty estimates.
- **Stacking ensemble:** Meta-learner on out-of-fold predictions from LightGBM + XGBoost + CatBoost.

15. Conclusion

This project demonstrates a complete end-to-end machine learning pipeline for flight delay prediction—from raw BTS data ingestion through feature engineering, model training, evaluation, and deployment as a production web application.

The system addresses temporal information leakage through multiple defence layers: a 14-column blacklist enforced at data loading, a chronological train/val/test split with assertion guards, K-Fold out-of-fold target encoding, and a three-stage leakage audit combining blacklist verification, high-correlation scanning, and SHAP importance analysis.

The 17-feature pipeline captures five distinct signal types—temporal (cyclic sin/cos), spatial (Haversine distance, 139 airports), categorical (K-Fold target encoding, $\lambda=20$), historical (route and airline delay aggregates), and calendar (holiday proximity ±2 days)—all derived exclusively from T-24h information.

LightGBM with Optuna-based TPE optimisation (50 trials, PR-AUC objective) is evaluated using PR-AUC as the primary metric, supplemented by ROC-AUC, F1, Brier score, and precision/recall at fixed operating points. SHAP TreeExplainer provides both global importance and per-prediction explanations.

The model is deployed as a Streamlit web application on Streamlit Community Cloud with an interactive 3D globe, SHAP waterfall chart, and airline-branded UI, as well as a FastAPI + React + Docker Compose alternative. Source code: **https://github.com/Paras-Dhatwalia/Flight-Delay-Predictor**.

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